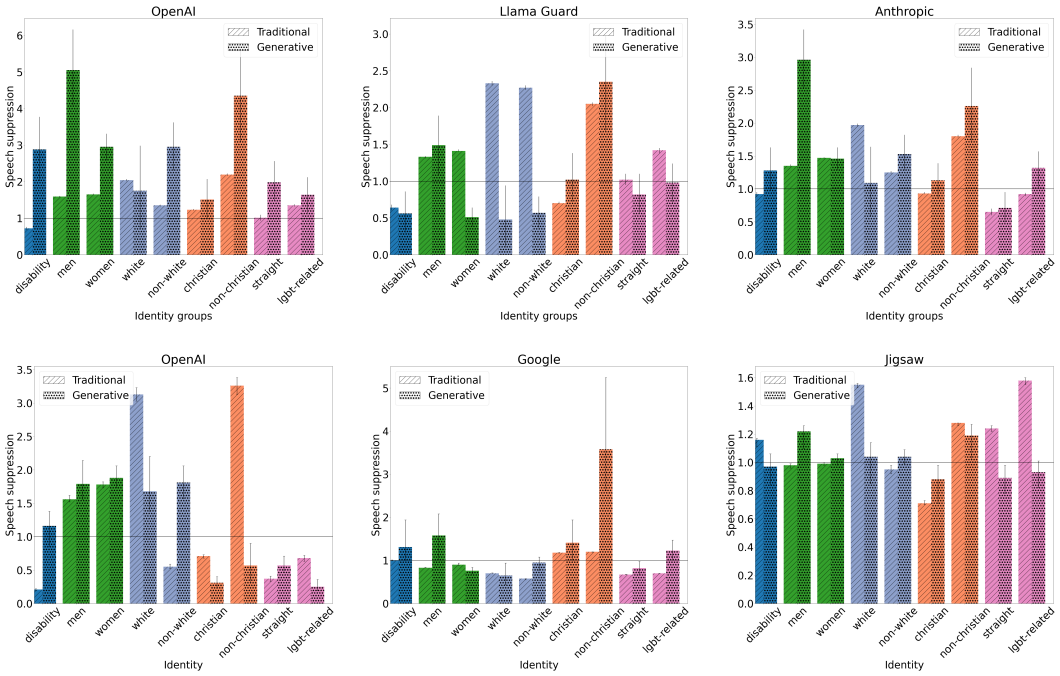

Figure 2: Identity-related speech suppression values for content moderation API results for flags (top row) and score values (bottom row) for both traditional and generative AI datasets. Error bars are the 95% confidence interval calculated with 1000 bootstrap samples drawn to the full dataset sizes. Values greater than 1.0 indicate that text related to that identity is flagged incorrectly more often (top row) or receives higher scores than usual (bottom row) for that dataset and API.

Variable	Anthropic	Llama Guard	OpenAI flag	OpenAI score	Google	Jigsaw
men	0.2837 ***	0.3508 ***	0.4636 ***	0.1642 ***	-0.0077 ***	0.0507 ***
women	0.6462 ***	0.7647 ***	0.6872 ***	0.2508 ***	0.0382 ***	-0.0861 ***
white	0.6089 ***	1.1958 ***	0.4398 ***	0.1367 ***	-0.0908 ***	0.1396 ***
non-white	0.1410 ***	0.8705 ***	0.3231 ***	0.0956 ***	-0.0980 ***	-0.1918 ***
christian	-0.1488 ***	-0.2610 ***	0.2231 ***	0.0718 ***	0.3053 ***	0.0084 ***
non-christian	1.3272 ***	1.6907 ***	1.1227 ***	0.4624 ***	0.0218 ***	0.0844 ***
straight	-0.5176 ***	0.0012	-0.0087	-0.0247 **	-0.0893 ***	0.0371 ***
lgbt	0.0085	0.6876 ***	0.5107 ***	0.1609 ***	-0.0305 ***	0.1388 ***
disability	0.0545 **	-0.1798 ***	-0.1525 ***	-0.0416 ***	0.1590 ***	0.0793 ***
GenAI	-2.2252 ***	-0.8198 ***	-3.0622 ***	-0.3921 **	-0.4043 ***	-0.0625 ***
has_slur	-0.6203 ***	-0.6222 ***	-0.0037	0.0165 ***	0.0494 ***	-0.0362 ***
word_length	0.1863 ***	-0.5018***	0.2069 ***	0.0511 ***	0.0442 ***	0.0061 ***

Table 3: Logistic regression results for Anthropic, Llama Guard, and OpenAI flag outputs (APIs outputting binary flags) predicting whether the API gives a false positive: flags on most identity tags are significantly more likely to be incorrectly flagged. Linear regression results for OpenAI score outputs, Google, and Jigsaw (APIs outputting continuous scores) on true negative text (content that should not be moderated) reflect greater speech suppression for identity-related text than other text: most identity tags are significantly more likely to receive higher (worse) content moderation scores. ** $p < 0.05$, *** $p < 0.005$.

5.3 Incorrectly suppressed text is often political speech or TV violence

To better understand the themes underpinning false positive responses from APIs, we conduct a qualitative review of these responses. We again focus on false positives (for the APIs that return flags) or incorrectly high scoring (for the APIs that return scores) responses. We sample 50 false positive or high scoring texts from each of the generative AI and traditional datasets for each API. For the Google and Jigsaw models, we take our sample from the 1500 highest flagging true negative instances. For each sampled instance, we hand code the instance into zero, one, or more of six categories. Since many models share categorization aims (see Fig.

	Traditional							Generative AI						
API	Pol.	Rel.	Iden.	Sex.	Hate	Viol.	None	Pol.	Rel	Iden.	Sex.	Hate	Viol.	None
Jigsaw	6	2	12	12	0	2	23	3	2	4	11	0	24	19
Anthropic	35	11	9	3	1	3	8	10	3	3	13	0	32	8
OpenAI	42	17	23	3	3	8	2	10	7	7	25	1	29	1
Llama Guard	26	16	18	5	1	4	14	2	2	2	3	0	11	34
Google	25	5	1	0	1	14	20	5	4	4	6	2	30	15

Table 4: The qualitative categorization of 50 sampled instances per API for both the traditional and generative AI datasets based on categories: politics (Pol.), religion (Rel.), identity bias (Iden.), sexual content (Sex.), hate, violence (Viol.), and no category (none).

1) regarding sexual content, violence, and targeted hate, we include these categories in our analysis, and add three additional groups based on our observations of prevalence in a pilot test: religion, politics, and identity bias. For religion and politics, we consider any mention of religious or political figures, systems, or ideologies to count towards their respective categories. We define a text as containing identity bias if it exhibits any biased speech against a named identity group, such as Muslim, LGBT, or Mexican people. Finally, we consider targeted hate to include mentions of violence, threats, or hate speech against specific identity groups.

Based on our qualitative coding, we find different trends for false positive results within generative AI and traditional datasets. On the traditional dataset, Anthropic, OpenAI, and Llama Guard all predominantly flag text regarding politics, religion, and identity bias, the content of which is often argumentative (see examples in Appx. Tables 16–18). Google also tends to flag political speech, though this is followed by high flagging rates for violence, not religious or identity biased speech. Jigsaw diverges from these trends in the traditional dataset by flagging uncategorized, identity bias, and sexual content the most; the model largely flags text which uses inappropriate language, even if not as an insult (see Appx. Table 16). For the generative dataset, we find that violence is the dominant category across APIs for false positive text, followed by sexual or uncategorized content. Many of these violent texts are from TV show or movie synopses labeled as `PG-13-ok` exhibiting violent themes such as shootings, action sequences, or paranormal activity (this finding matches that of previous work on OpenAI in the context of TV shows [46]). Notably, Llama Guard predominantly flags content which does not fall under a qualitative assessment category. Upon reading these texts, we find that the model is often flagging TV and movie descriptions which appear largely benign (see Appendix Table 18).

5.4 Identity groups are incorrectly flagged based on stereotypes and text associations

In order to identify why identity-related speech suffers from suppression on both the traditional and generative AI datasets, we further consider the speech suppression scores per-identity and per-category across the traditional and generative AI datasets (see Appx. Figures 7–10). We find that text pertaining to different identity groups are flagged for different reasons. For example, when the Google API is applied to the generative AI dataset, disability-related content receives a high speech suppression score and is flagged for "Health" and non-Christian content is flagged for "Religion & Belief." On the Llama Guard API we find that generative AI content related to men is incorrectly flagged for "Violence & Hate" and for "Sexual Content." From the literature (see Sec. 2) we know that different identity groups have associations in most large language models, often based on stereotypes, that must be purposefully unbiased if not desired.

To examine these further, we consider the speech suppression scores for the identity categories of disability, sexual orientation, and religion across APIs that return per-category scores or flags (OpenAI, Llama, and Google). We consider categories that are relatively consistent in flagging aim across API and for which we anticipate a stereotypical or other textual association with the identities based on the literature. In Figure 3 we see that across APIs: disability-related content is more likely to be flagged for self-harm than other content; straight and LGBT content is more likely to be flagged for sexual content than non-identity-related content, and; non-Christian content is more likely to be flagged for hate than Christian or non-identity-related content.

These results are somewhat expected based on known stereotypes, but also provided needed specificity to the literature. LGBT content is stereotypically associated with inappropriate sexual content, but we find here that both straight and LGBT content are incorrectly flagged by these systems as sexual. It's known that Muslim content is commonly associated with violence by large language models [1], but these results additionally provide initial evidence that non-Christian content is heavily associated with hateful speech, leading speech related to these groups to be incorrectly flagged as hateful. While disability content is known to be identified with negative sentiment [68], examining the per-category flags also allows us to identify the association between disability and speech suppression related to self-harm. Much of the literature on bias focuses on a binary identification of toxic content (see, e.g., [14, 24]), and thus may miss some of this per-identity and per-category nuance. Examining these stereotypes and text associations in this fine-grained way may allow future avenues for bias reduction to be identified.

Additionally, to better understand the extent to which these stereotypes and negative textual associations that lead to incorrect flagging behavior are based on the identity terms themselves or the broader context of the examined text, we turn to the speech suppression results on the Jigsaw Bias dataset (see Table 5). Recall

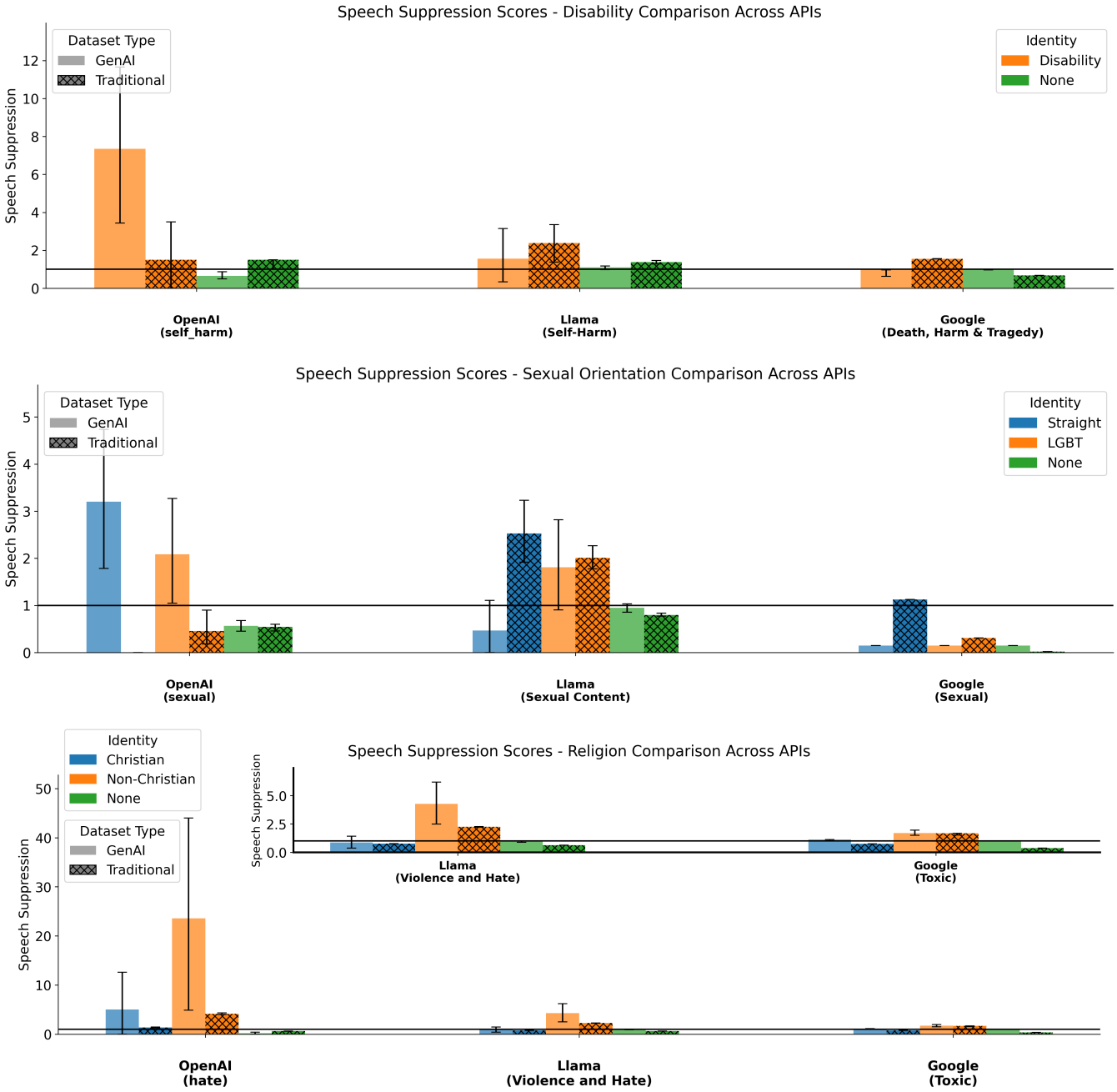

Figure 3: Comparison of speech suppression scores across different APIs and identity categories (disability, sexual orientation, and religion). Error bars represent 95% confidence intervals based on 1000 bootstrap samples. The religion chart includes a zoomed in version of two of the APIs for clearer comparison.

that this dataset is created in a template-based manner, with different explicit identity terms substituted into the same phrases. This allows us to see what the speech suppression scores are like across identity group and API in cases where the broader context of the text has been removed and suppression is likely due to the identity terms themselves. We see that all APIs still suffer from speech suppression for some identity groups on this dataset. Interestingly, OpenAI's content moderation endpoint was directly trained using this template-like approach [48]. We see that while this brought the speech suppression for most identity groups on the dataset to essentially 0 for OpenAI's flag-based results, this was notably unsuccessful for the LGBT and straight groups. Across APIs, this